```
from google.colab import files
uploaded = files.upload()

import cv2
import numpy as np
import matplotlib.pyplot as plt

img = cv2.imread(list(uploaded.keys())[0])
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

h, w = img_rgb.shape[:2]

src_pts = np.array([
    [50, 50],
    [w-50, 50],
    [w-50, h-50],
    [50, h-50]
], dtype=np.float32)

dst_pts = np.array([
    [0, 0],
    [w, 0],
    [w-100, h],
    [100, h]
], dtype=np.float32)

def compute_homography_dlt(src, dst):
    A = []
    for i in range(len(src)):
        x, y = src[i]
        u, v = dst[i]

        A.append([-x, -y, -1, 0, 0, 0, x*u, y*u, u])
        A.append([0, 0, 0, -x, -y, -1, x*v, y*v, v])

    A = np.array(A)

    U, S, Vt = np.linalg.svd(A)
    H = Vt[-1].reshape(3, 3)

    return H / H[2, 2]

# Compute homography matrix using DLT
H_dlt = compute_homography_dlt(src_pts, dst_pts)

# -------------------------------
# Step 3: Apply Transformation
# -------------------------------
dlt_transformed = cv2.warpPerspective(img_rgb, H_dlt, (w, h))
```

```
# ------------------------------
# Step 4: Display Output
# ------------------------------
plt.figure(figsize=(12,5))

plt.subplot(1,2,1)
plt.title("Original Image")
plt.imshow(img_rgb)
plt.axis('off')

plt.subplot(1,2,2)
plt.title("DLT Transformed Image")
plt.imshow(dlt_transformed)
plt.axis('off')

plt.show()
```

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```
Saving dragon.png to dragon (1).png
```